Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
9	(a)	(i)		all **three** needed chloride – white bromide – cream (accept off-white) yellow – iodide	1			1		1
		(ii)		$\Delta_f H^\theta$ (AgBr) = 105.6 + (½ × –243.1) – 84.4 (1) $\Delta_f H^\theta$ (AgBr) = –100.35 (1) AgCl more stable than AgBr (ecf possible from values) (1) as standard enthalpy of formation is more negative (1)		2	2	4	1	
	(b)	(i)	I	$Cu + 2Ag^+ \rightarrow Cu^{2+} + 2Ag$		1		1		
			II	E^θ (Cu) must be more negative/less positive than +0.80V (1) award (1) for either of following reasons reaction is feasible when EMF is positive reaction is feasible when reduction half-cell is more positive than oxidation half-cell			2	2		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
		(ii)	I	award (2) for correct answer $E^{\theta} = 0.80 - 0.93 = -0.13$ V if incorrect award (1) for use of 0.93 V		1	1	2	1	
			II	as concentration of Ag^+ decreases the $Ag^+ + e^- \rightleftharpoons Ag$ equilibrium shifts to the left (1) this makes E for this reaction less positive (1) EMF of the cell is $E_{Ag} - E_{Pb}$ so the value becomes less positive (1)			3	3		
				Question 9 total	**1**	**4**	**8**	**13**	**2**	**1**